CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

5.1 Energy Conservation

Syllabus Section: 5. Work, Energy & Power **Paper Type:** Multiple Choice (Paper 1) **Total:** 34 Questions (34 Marks)

Question 1 Easy [1 mark]

Which of the following is a statement of the principle of conservation of energy?

A energy is the product of force and distance

B energy cannot be created, destroyed or transformed

C the total energy of an isolated system is constant

D the power transmitted by a system is proportional to the rate of transfer of energy

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The Principle of Conservation of Energy states that energy cannot be created or destroyed, it can only be transferred from one form to another
- In other words, the total energy of an **isolated** system remains constant
- An isolated system is defined as a system that is closed off from its surroundings and this means that a system always has the same amount of energy, unless it's added from an external source
- Identifying the system is the most difficult part of conservation questions. Let's take a look at an example of a ball tossed up in the air:
- *During the throw:*
  - Chemical energy from your muscles → kinetic energy of the ball
- *As the ball reaches its maximum height:*
  - Kinetic energy of the ball → gravitational potential energy of the ball
- *As the ball falls down:*
  - Gravitational potential energy of the ball → kinetic energy of the ball
- The isolated system in this scenario is simply the ball, the muscles in the person's arm, and the Earth's gravitational field. It's not necessary to consider the chemical energy from the food the person ate or the gravitational fields of other bodies such as the sun or moon, for example

A & D are incorrect because these describe equations to work out work done and power, but they don't describe conservation of energy

B is incorrect because this is close to the classic definition 'energy cannot be created or destroyed; it can only be transferred from one form to another' but if you look closely you will see this definition states that energy cannot be transformed which is incorrect

Question 2 Easy [1 mark]

Which expression **defines** power?

A force × displacement in the direction of the force

B force × velocity

C work done ÷ time taken

D work done × time taken

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Power is defined as the rate of transfer of energy or work done
- In equation form this can be written as
  - Power, $P = \frac{energy}{time}$ or $P = \frac{work\ \ done}{time}$

A is incorrect because this equation describes how to calculate work done

B is incorrect because this is a correct formula for power but it derived from $Power = \frac{work\ \ done}{time}$

D is incorrect because this is not a correct formula, it doesn't have a physical equivalent

Question 3 Easy [1 mark]

The total energy input E_{in} in a process is partly transferred to useful energy output U, and partly to energy that is wasted W

What is the efficiency of the process?

A $\frac{U}{W} \times 100\%$

B $\frac{W}{E_{in}} \times 100\%$

C $\frac{U}{E_{in}} \times 100\%$

D $\frac{U+W}{E_{in}} \times 100\%$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The efficiency of a system is the ratio of the energy used in the process to the total energy input in the system
- Efficiency $= \frac{useful\ \ energy\ \ output}{total\ \ energy\ \ input}$
- Useful energy output = U
- Total energy input = E_{in}
    - % Efficiency $= \frac{U}{E_{in}} \times 100\%$

Question 4 Easy [1 mark]

A boat moving at constant speed *v* through still water experiences a total frictional drag *F*

What is the power developed by the boat?

A $\frac{1}{2}Fv$

B Fv

C $\frac{1}{2}Fv^2$

D Fv^2

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Power is defined as the rate of transfer of energy or work done
- In equation form this can be written as
  - Power, $P = \frac{energy}{time}$ or $P = \frac{work\ \ done}{time}$
- Work done = Force × displacement = $F \times s$
  - $P = \frac{work\ \ done}{time} = \frac{Fs}{t} = Fv$
  - Where speed, $v = \frac{s}{t}$
- Therefore, $P = Fv$

Question 5 Easy [1 mark]

What is the average power output of a laser that can deliver 0.20 J of energy in 10 ns?

A 2 nW

B 20 mW

C 200 kW

D 20 MW

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Power is defined as the rate of transfer of energy or work done
- This can be written as $P = \frac{energy}{time}$
- Time = 10 ns = 10×10^{-9} s
  - $P = \frac{0.20}{10 \times 10^{-9}} = 2 \times 10^{7}$ W
- 1 MW = 1×10^{6} W
  - $P = 2 \times 10^{7}$ W $= 20$ MW
- Therefore, the average power output of the laser 20 MW

Question 6 Easy [1 mark]

A transformer has the following input and output

	Potential difference / V	Current / A
Input	11 000	28
Output	240	1200

What is the efficiency of the transformer?

A 0.94 %

B 1.0 %

C 11 %

D 94 %

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The efficiency of a transformer is the ratio of the power output of the process to the total power input to the system
- Efficiency = $\frac{useful\ \ power\ \ output}{total\ \ power\ \ input}$
- Power can be calculated using : $P = VI$
- Useful power output $= 240 \times 1200 = 288000$
- Total power input = $11\ \ 000 \times 28 = 308000$
  - % Efficiency = $\frac{useful\ \ power\ \ output}{total\ \ power\ \ input} \times 100\%$
  - % Efficiency = $\frac{288000}{308000} \times 100\ \% = 93.5\%$
- Therefore, the efficiency of the transformer is 94%

Question 7 Easy [1 mark]

Which operation involves the greatest mean power?

A a car moving against a resistive force of 0.4 kN at a constant speed of 20 m s^{-1}

B a crane lifting a weight of 3 kN at a speed of 2 m s^{-1}

C a crane lifting a weight of 5 kN at a speed of 1 m s^{-1}

D a weight being pulled across a horizontal surface at a speed of 6 m s^{-1} against a frictional force of 1.5 kN

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Average, or mean, power is defined as the rate of transfer of energy or work done
- In equation form, this can be written as
  - Power, $P = \frac{\text{energy}}{\text{time}}$ or $P = \frac{\text{work done}}{\text{time}}$
- Work done = Force × displacement = Fs
  - $P = \frac{\text{work done}}{\text{time}} = \frac{Fs}{t} = Fv$
- In scenario D :
  - Force, F = 1.5 kN
  - Speed, v = 6 m s^{-1}
- $P = Fv = 1.5 \times 6 = 9 \text{ kW}$
- This operation gives the largest mean power, see the incorrect table for the other calculations

A is incorrect because $P = Fv = 0.4 \times 20 = 8 \text{ kW}$

B is incorrect because $P = Fv = 3 \times 2 = 6 \text{ kW}$

C is incorrect because $P = Fv = 5 \times 1 = 5 \text{ kW}$

Question 8 Easy [1 mark]

An electric motor is required to produce 120 W of mechanical output power. The efficiency of the motor is 80%

Which row is correct?

	Electrical power input to motor / W	Waste heat output from motor / W
A	120	24
B	120	96
C	150	30
D	150	120

A

B

C

D

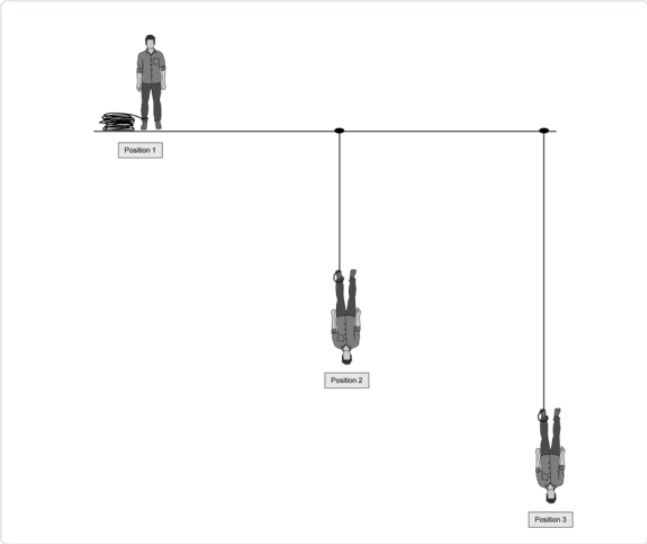
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Efficiency $= \frac{useful\ \ mechanical\ \ output\ \ power}{total\ \ electrical\ \ power} = 80\%$
- Useful mechanical output power = 120 W
- Input power $= \frac{useful\ \ mechanical\ \ output\ \ power}{efficiency} = \frac{120}{0.8} = 150\ \ \text{W}$
- Waste heat power = input – output = 150 – 120 = 30 W

Question 9 Easy [1 mark]

When bungee jumping, a student starts with maximum gravitational potential energy (position 1), then falls freely until the rope fully unwinds (position 2), after which the rope starts to stretch until the lowest point of the jump is reached (position 3)



What are the kinetic and elastic potential energies at position 3?

	Kinetic energy	Elastic potential energy
A	maximum	maximum
B	maximum	minimum
C	minimum	maximum
D	minimum	minimum

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- From the conservation of energy, the sum of energies at position 2 or 3 should be equal to the initial amount of gravitational potential energy the student possesses at position 1
- At position 2, the rope fully unwinds but it does not stretch. So, it does not have elastic potential energy. The kinetic energy is maximum at position 2
- At position 3, the rope is stretched. So, the elastic potential energy will reach a maximum
- At this position, the lowest point of the jump is reached, so, the student will not move any further, thus, the kinetic energy will be at a minimum

Question 10 Easy [1 mark]

The kinetic energy of a moving vehicle with mass 1000 kg is 4.5×10^5 J. It brakes with a total constant braking force of 6000 N

What will be its stopping distance?

A 37 m

B 75 m

C 150 m

D 300 m

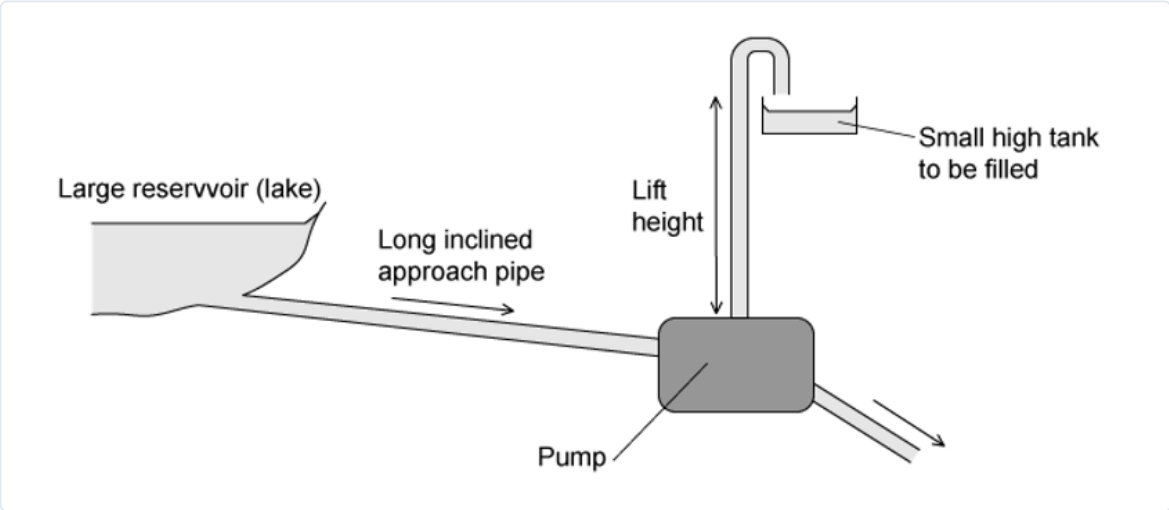
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Work done to stop vehicle = kinetic energy = 4.5×10^5 J
- Work done = Force × Distance
- Distance $= \frac{work\ \ done}{force} = \frac{4.5 \times 10^5}{6000} = 75 \quad \mathrm{m}$

Question 11 Medium [1 mark]

The diagram shows a pump called a hydraulic ram



In one such pump the long approach pipe holds 500 kg of water. A valve shuts when the speed of this water reaches 2.0 m s^{-1} and the kinetic energy of this water is used to lift a small quantity of water by a height of 15 m.

The efficiency of the pump is 10 %.

Which mass of water could be lifted 15 m?

A 0.15 kg

B 0.68 kg

C 1.5 kg

D 6.8 kg

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In this question we need to consider conservation of energy since energy is being transformed form one form into another
- The kinetic energy of the water is being converted into gravitational potential energy of the water
- However, since the efficiency of the pump is only 10%, only 10% of the kinetic energy is converted into GPE
- 10% of KE of the water = GPE of water lifted
  - $0.1 \times \frac{1}{2}mv^2 = m_w g \Delta h$

- Where m_w is the mass of the water lifted
  - $0.1 \times \frac{1}{2} \times 500 \times 2^2 = m_w \times 9.81 \times 15$
  - $m_w = \frac{0.1 \times 500 \times 4}{2 \times 9.81 \times 15} = 0.68 \quad \text{kg}$
- Therefore, 0.68 kg of water could be lifted by 15 m

Question 12 Medium [1 mark]

Trains supply coal to a power station. The table below gives quantities describing the operation of the power station.

	Symbol	Unit
power station output	*P*	W
number of trains per day	*N*	
mass of coal on a train	*M*	kg
energy from 1 kg of coal	*J*	J
number of seconds in one day	*S*	s

Which expression gives efficiency of the power station?

A $\frac{PS}{NMJ}$

B $\frac{PSN}{MJ}$

C $\frac{NMJ}{PS}$

D $\frac{NM}{PSJ}$

Mark Scheme & Explanation:

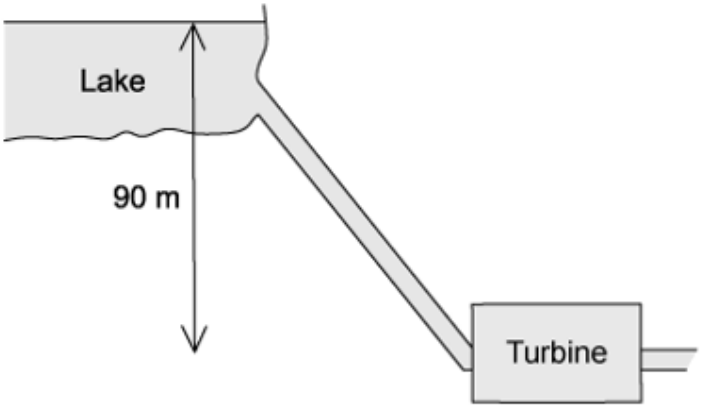
*The correct answer is **A** because:*

- Efficiency is a quantity that describes a ratio of two similar appropriate quantities, so, the numerator and the denominator must have the same units
- In this scenario, we can give the efficiency in terms of input and output energy per day, we will need to use the following equations:
- $Efficiency = \frac{output\ \ energy\ \ per\ \ day}{total\ \ energy\ \ available\ \ per\ \ day}$
- $Power = \frac{energy}{time}$
- $Energy = Power \times Time$

    - Output energy = Power station output × Number of seconds in one day
    - Output energy = PS
    - Total energy available per day = Number of trains per day × Mass of coal on a train × Energy from 1 kg of coal
    - Total energy available per day = NMJ
- Efficiency $\frac{output\ \ energy\ \ per\ \ day}{total\ \ energy\ \ available\ \ per\ \ day} = \frac{PS}{NMJ}$

Question 13 Medium [1 mark]

Water flows from a lake into a turbine that is a vertical distance of 90 m below the lake, as shown



The mass flow rate of the water is 2400 kg min^{-1}. The turbine has an efficiency of 75 %.

What is the output power of the turbine?

A 26 kW

B 35 kW

C 1.6 MW

D 2.1 MW

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Power is the rate of transfer of energy
  - $Power = \frac{energy}{time}$
- Water in the lake possesses gravitational potential energy
  - $Power = \frac{\Delta E_p}{t} = \frac{mg\Delta h}{t} = \frac{m}{t}g\Delta h$
- The mass flow rate of the water $\frac{m}{t}$ is 2400 kg min^{-1}
- This means that every minute, 2400 kg of water flows through the turbine
  - In 60 s, 2400 kg of water flows

- Therefore, in 1 s, $\frac{2400}{60} = 40 \quad \text{kg}$ of water flows

- The mass flow rate is equal to $\frac{m}{t}$ = 40 kg s^{-1}
  - Power of water = $\frac{m}{t} g \Delta h = 40 \times 9 . 81 \times 90 = 35 \quad 316 \text{ W}$
- This is the maximum power available from the turbine, however, the turbine has an efficiency of 75%
  - Power output of the turbine = $0 . 75 \times 35 \quad 316 = 26 \quad 487 = 26 \quad \text{kW}$

Question 14 Medium [1 mark]

Initially, four identical uniform blocks, each of mass *m* and thickness *h*, are spread on a table.



How much work is done on the blocks in stacking them on top of one another?

A 3 mgh

B 6 mgh

C 8 mgh

D 10 mgh

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- To stack the blocks on top of each other, we only need to lift three of them, as the bottom block is already on the table
- When on the ground, a block has zero gravitational potential energy
- When at a height *x* above the ground, the gravitational potential energy of a block is *mgx*
- So, the work that needs to be done to lift a block from the ground to a height *x* is *mgx*, where *x* will equal *h*, 2*h*, 3*h*
    - The first block is lifted by a height *h*, so, work done = *mgh*
    - The second block is lifted by a height 2 *h*, so, work done = 2 *mgh*
    - The third block is lifted by a height 3 *h*, so, work done = 3 *mgh*
- Total work done = *mgh* + 2 *mgh* + 3 *mgh* = 6 *mgh*

Question 15 Medium [1 mark]

The first column in the table gives four examples of work being done. The second column gives more detail of the action.

Which row is **not** correct?

	Example	Detail
A	a girl dives from a diving board into a swimming pool	work is done by the girl against gravity as she falls
B	a man pushes a car along a level road	work is done by the man against friction
C	an electron is accelerated towards a positively-charged plate	work is done on the electron by the electric field of the plate
D	a piston is pushed outwards as a gas expands	work is done on the atmosphere by the gas

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Work done = Force × displacement in direction of the force
- In situation A, a girl dives from a diving board into a swimming pool
- The direction of the gravitational force is downwards
- The gravitational force acts downward on the girl and the girl moves downward during the fall
- So work is done by the gravitational force, not against it

B is correct because a man pushes a car along a level road:

- direction of (electric) force = horizontal ✓
- direction of motion = horizontal ✓

C is correct because an electron is accelerated towards a positively-charged plate:

- direction of (electric) force = towards positively charged plate ✓

- direction of motion = electron moves towards the positively charged plate ✓

D is correct because a piston is pushed outwards as a gas expands:

- direction of force = on piston (the gas in the atmosphere exerts force on the surface of the piston) ✓
- direction of motion = piston moves outwards ✓

Question 16 Medium [1 mark]

The data below are taken from a test of a petrol engine for a motor car.

power output	150 kW
fuel consumption	20 litres per hour
energy content of fuel	40 MJ per litre

Which expression will evaluate the efficiency of the engine?

A $\frac{150 \times 10^3}{40 \times 10^6 \times 20 \times 60 \times 60}$

B $\frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$

C $\frac{150 \times 10^{3\times} \ 40 \times 10^6 \times 20}{60 \times 60}$

D $\frac{150 \times 10^3 \times 20}{40 \times 10^6 \times 60 \times 60}$

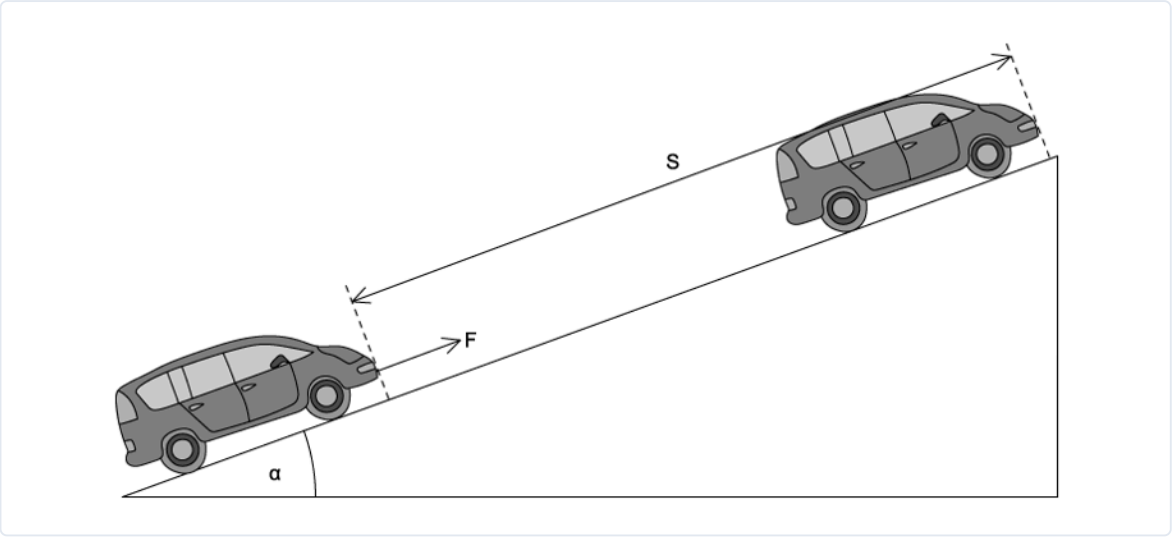
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Efficiency = $\frac{useful\ \ output\ \ power}{total\ \ power\ \ input}$
- We know the power output is = 150 kW = 150 × 10^3 W
- Power input = $20 \times 40MJ = 20 \times 40 \times 10^6$
- We must convert the fuel consumption into litres per second
  - In one hour, the fuel consumed is 20 litres
  - In one minute, the fuel consumed is $\frac{20}{60}$ litres
  - In one second, the fuel consumed is $\frac{20}{60 \times 60}$ litres
- So, power input = $\frac{20 \times 40 \times 10^6}{60 \times 60}$
- Efficiency = $\frac{output\ \ power}{power\ \ input} = \frac{150 \times 10^3}{\frac{20 \times 40 \times 10^6}{60 \times 60}} = \frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$

Question 17 Medium [1 mark]

A constant force F acting on a car of mass m moves it up a slope through a distance s at constant velocity v. The angle of the slope to the horizontal is α.



Which expression gives the efficiency of the process?

A $\frac{mgs \sin \alpha}{Fv}$

B $\frac{mv}{Fs}$

C $\frac{mv^2}{2Fs}$

D $\frac{mg \sin \alpha}{F}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

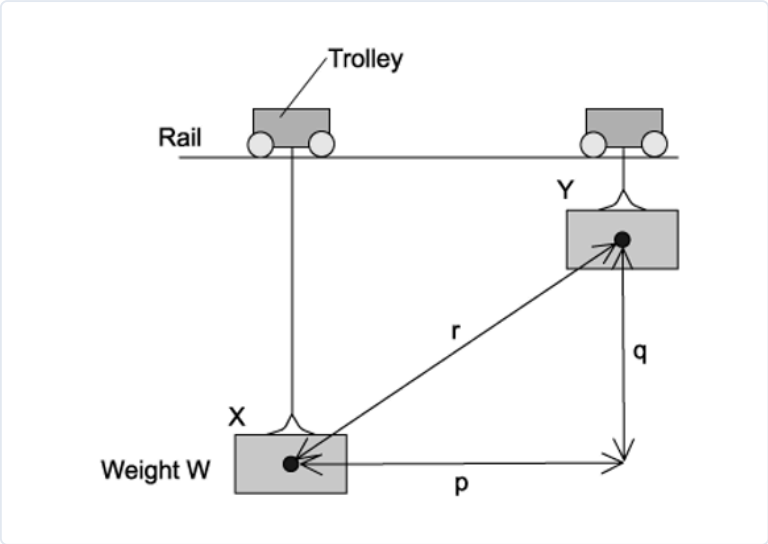
- Efficiency is given by
  - efficiency = $\frac{\text{useful energy output}}{\text{total energy input}}$
- The total energy input is equal to the work done by the car, which is given by
  - $W = Fs$
- For constant velocity, the resultant force is zero
- So, the force required to move the car up the slope is equal to the component of weight which acts along the slope
  - Component of weight against $F = mg \sin \alpha$

- Work done to move car a distance $s = mg \sin \alpha s$

- Efficiency $= \frac{\text{useful energy output}}{\text{total energy input}} = \frac{mg \sin \alpha s}{Fs} = \frac{mg \sin \alpha}{F}$

Question 18 Medium [1 mark]

A weight *W* hangs from a trolley that runs along a rail. The trolley moves horizontally through a distance *p* and simultaneously raises the weight through a height *q*.



As a result, the weight moves through a distance *r* from X to Y. It starts and finishes at rest.

How much work is done on the weight during this process?

A *Wp*

B *W(p + q)*

C *Wq*

D *Wr*

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Work done = Force applied × displacement moved in the direction of force
- In this case the force acting on the mass is the tension in the string holding the mass which is equal to the weight *W* in this case
- The distance moved by the weight in direction of this force is *q*
    - So, work done $= W_q$

A & D are incorrect because if the force had also been given along the track then the work done could be represented as F_p *or* F_r or , but since we do know the force in the vertical direction is weight *W*, then we can

only specify the work done by the weight

B is incorrect because these displacements are in different directions, whereas work done requires the displacement to be in the same direction as the force

Question 19 Medium [1 mark]

A bungee jumper has 24 kJ of gravitational potential energy at the top of his jump. He is attached to an elastic rope which starts to stretch after a short time of free fall. Assume that energy loss through air resistance is negligible.

	GPE / kJ	EPE / kJ	KE / kJ
Top	24	0	0
Bottom	0	24	0

What are the possible values when the jumper is half-way down?

	GPE / kJ	EPE / kJ	KE / kJ
A	12	10	2
B	12	8	4
C	8	8	8
D	12	2	10

A

B

C

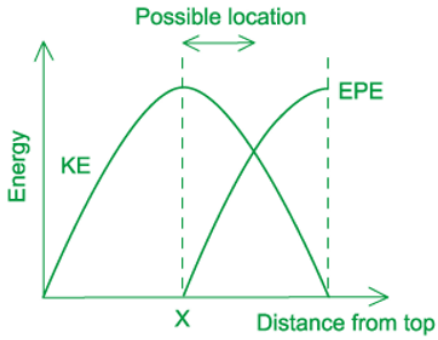
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

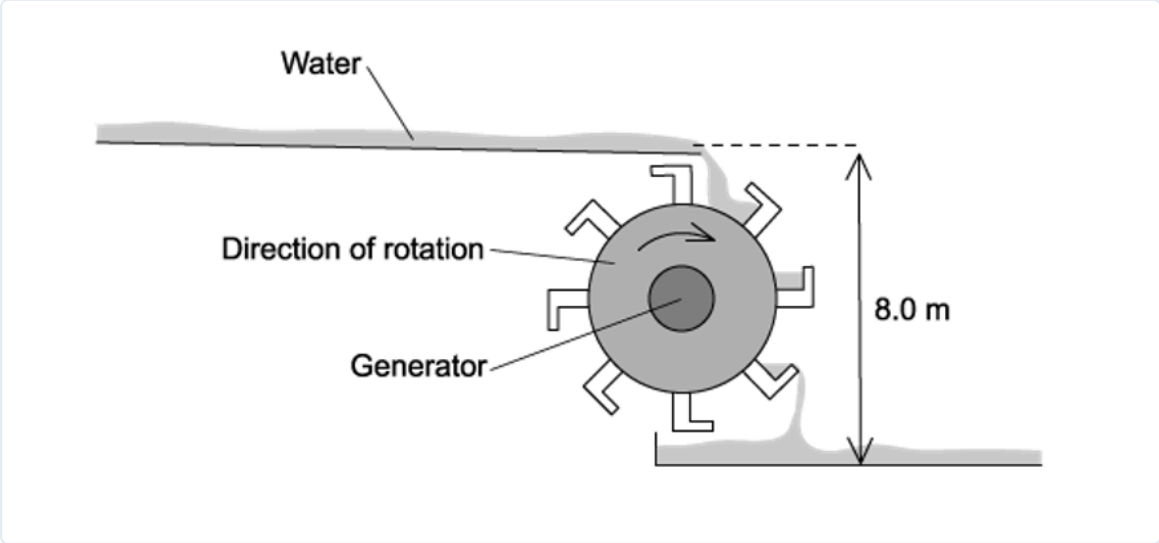
- From the conservation of energy, the sum of energies halfway and at the bottom of the jump should be equal to the initial amount of gravitational potential energy the student possesses at the beginning of the jump
- When the jumper is halfway down, the rope will unwind fully but it will not have stretched yet. So, it will have a small elastic potential energy
- The kinetic energy will reach a maximum halfway down

- The graph below might give you a clearer image
- Point X shows the point where the jumper starts to be restricted by the pull of the rope
- Before X, the jumper undergoes free fall and hence the speed and kinetic energy increases
- After X, the elastic potential energy of the elastic rope starts increasing – this corresponds to a drop in the kinetic energy of the jumper, according to the principle of conservation of energy
- When the jumper is halfway down, their possible location is shown on the graph, it can clearly be seen that the kinetic energy is greater than the elastic potential energy at this point, making D the correct answer



Question 20 Medium [1 mark]

The diagram shows the design of a water wheel which drives a generator to produce electrical energy. The flow rate of the water is 200 kg s^{-1}. The generator supplies a current of 32 A at a voltage of 230 V.



Ignoring any changes in kinetic energy of the water, what is the efficiency of the system?

A 14 %

B 16 %

C 22 %

D 47 %

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Efficiency $= \frac{power\ \ output}{power\ \ input}$
- Power output $= IV$
- Power output = $32 \times 230 = 7360$
- Water in the top section possesses gravitational potential energy
  - $Power = \frac{\Delta E_p}{t} = \frac{mg\Delta h}{t} = \frac{m}{t} g\Delta h$
- The mass flow rate is equal to $\frac{m}{t} = 200 \quad \text{kg} \quad \text{s}^{-1}$
  - Power input of water = $\frac{m}{t} g\Delta h = 200 \times 9.81 \times 8 = 15 \quad 696 \quad \text{W}$

- % Efficiency = $\frac{power\ \ output}{power\ \ input} \times 100\,\% = \frac{7360}{15\ \ 696} \times 100 = 46.9\%$
- Therefore, the efficiency of the system is 47%

Question 21 Medium [1 mark]

A conveyor belt is driven at velocity V by a motor. Sand drops vertically on to the belt at a rate of $\mathrm{m\ \ kg\ \ s^{-1}}$.

What is the additional power needed to keep the conveyor belt moving at a steady speed when the sand starts to fall on it ?

A $\frac{1}{2}mv$

B mv

C $\frac{1}{2}mv^2$

D mv^2

Mark Scheme & Explanation:

The correct answer is D because:

- Power is equal to = $P = \frac{E}{t} = \frac{Fs}{t} = Fv$
- Force is defined as the rate of change of momentum, so, we first need to find an expression for force in order to find the power
- sand drops vertically on to the belt at a rate of $\mathrm{m\ \ kg\ \ s^{-1}}$
  - Therefore, the rate at which the sand falls is = $\frac{\Delta mass}{time} = \frac{\Delta m}{t} = \mathrm{m\ \ kg\ \ s^{-1}}$
- Next, consider the momentum : $p = Mv$
- Rate of change of momentum = $\frac{\Delta p}{t} = \frac{Mv}{t} = \frac{\Delta M}{t}v = mv$
  - Therefore, in one second, the momentum of mass m of sand increases from Zero to mv
- Recall the force is defined as the rate of change of momentum, so the force involved is mv
- Force = rate of change of momentum = mv
  - so, Power = $Fv = mv \times v = mv^2$

Question 22 Medium [1 mark]

The force resisting the motion of a car is taken as being proportional to the square of the car's speed. The magnitude of the force at a speed of $20 \quad \text{m} \quad \text{s}^{-1}$ is 800 N.

What effective power is required from the car's engine to maintain a steady speed of $40 \quad \text{m} \quad \text{s}^{-1}$?

A 32 kW

B 64 kW

C 128 kW

D 512 kW

Mark Scheme & Explanation:

The correct answer is C because:

- We are told the resistive force F is proportional to v^2
  - $F \quad \alpha \quad v^2$
  - $F = kv^2$ where k is constant
- When F=800 N, V = 20 m s^{-1}
  - $k = \frac{F}{v^2} = \frac{800}{20^2} = 2$
- so, $F = 2v^2$
- Power output is equal to $P = Fv$
- Since $F = 2v^2$ we have :
  - Output power $P = Fv = 2v^2 \times v = 2v^3$
- So, when $V = 40 \quad \text{m} \quad \text{s}^{-1}$
  - Output power $P = 2v^3 = 2 \times 40^3 = 128 \quad 000\text{W}$
- Therefore, the effective power output is 128 kW

Question 23 Medium [1 mark]

A mass is raised vertically. In a time t, the increase in its gravitational potential energy is E_p and the increase i its kinetic energy is E_k .

what is the average power input to the mass?

A $E_p - E_k t$

B $E_p + E_K t$

C $\frac{E_p - E_k}{t}$

D $\frac{E_p + E_k}{t}$

Mark Scheme & Explanation:

*The correct answer is **D** because :*

- From the question we are told :
  - Increase in gravitational potential energy =E_p
  - Increase in kinetic energy $= E_k$
  - Time taken $= t$
- Let the total gravitational energy be equal to E
- We know that gravitational energy is the sum of potential energy and kinetic energy so
  - $E = E_p + E_k$
- Let P be the power input of the mass
- Power is equal to $P = \frac{E}{t}$
  - Therefore : $P = \frac{E}{t} = \frac{E_p + E_k}{t}$

Question 24 Hard [1 mark]

A turbine at a hydroelectric power station is situated 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

The overall efficiency of the turbine and generator system is 90 %.

What is the output power of the power station? (The density of water is 1000 kg m^{-3})

A 0.15 MW

B 1.5 MW

C 1.7 MW

D 90 MW

Mark Scheme & Explanation:

*The correct answer is **B** because:*

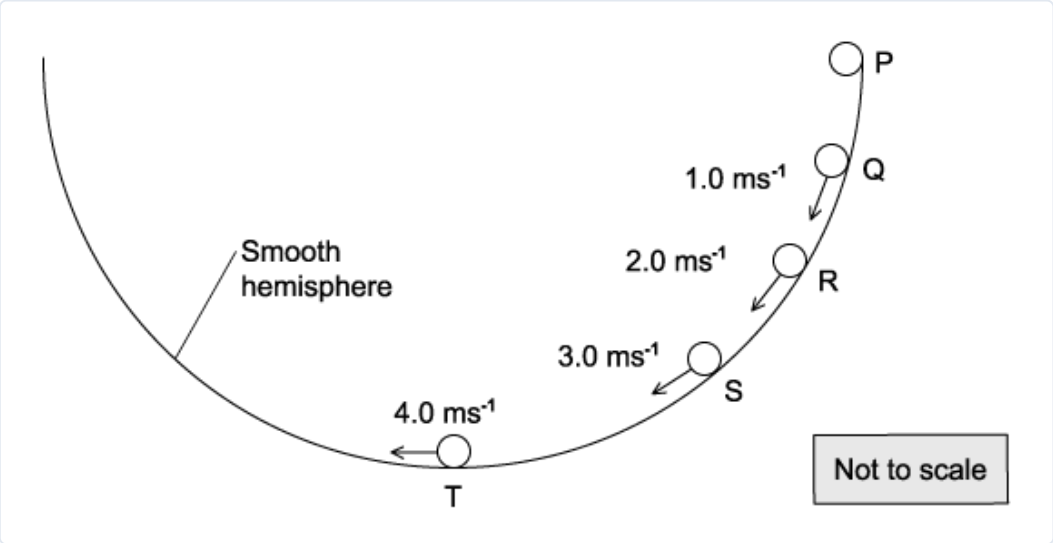
- At the top of the lake, the water has gravitational potential energy as it is above the ground
- As it passes through the turbine, the energy is converted to electrical energy, however, the system is only 90% efficient, so, not all of the gravitational potential energy becomes electrical energy
  - GPE of water $\Delta E_p = mg\Delta h$
  - Input Power = $\frac{E}{t} = \frac{\Delta E_p}{t} = \frac{mg\Delta h}{t}$
- The water passes through the turbine at a rate of 340 m^3 per minute. This quantity represents the volume flow rate and has unit m^3 per unit or $\frac{\Delta V}{t}$
- Volume flow rate = $\frac{\Delta V}{t} = 340 \quad m^3\,min^{-1}$
- We need to convert this into SI unit
  - In 1 minute, a volume of 340 m^3 passes through the turbine
  - In 1 second, a volume of $\frac{340}{60} = 5.67 \quad m^3$ passes through the turbine
- So, the volume flow rate is $\frac{\Delta V}{t} = 5.67 \quad m^3\,s^{-1}$
- The formula for the input power above requires the mass, not the volume. We need to convert the volume flow rate into mass flow water or $\frac{\Delta m}{t}$
- Density is equal to $\rho = \frac{mass}{volume} = \frac{m}{V}$, so , $\Delta m = \rho V$
- We must divide by time to get the equivalent mass flow rate :
  - $\frac{\Delta m}{t} = \frac{\rho V}{t} = 1000 \times 5.67 = 5670 \quad kg \quad s^{-1}$

    - Input Power = $\frac{mg\Delta h}{t} = \frac{\Delta m}{t} \quad g\Delta h$
    - Input Power $5670 \times 9.81 \times 30 = 1.67 \times 10^6 \quad \text{W} = 1.67 \quad \text{MW}$
- The overall efficiency of the system is 90%
- Therefore, output power = 0.90 × 1.67 = 1.5 MW

Question 25 Hard [1 mark]

A small mass is placed at point P on the inside surface of a smooth hemisphere. It is then released from rest. When it reaches the lowest point T, its speed is 4.0 m s^{-1}.

The diagram (not to scale) shows the speed of the mass at other points Q, R and S as it slides down. Air resistance is negligible.



The mass loses potential energy *E* in falling from P to T.

At which point has the mass lost potential energy $\frac{E}{4}$?

A Q

B R

C S

D none of these

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- From conservation of energy:
- The potential energy (PE) lost in falling from P to T = Kinetic energy (KE) of mass at T
  - $E = \frac{1}{4}m4^2 = 8 \quad m$
- Since the mass is at rest at P, its KE is zero and its PE is at a maximum

- Since T is the lowest point, PE is zero here
- So, the total energy of the mass in the system is *E*
- In general, PE lost at a point = KE at that point
- From above, *E* = 8*m*, so, a PE loss of $\frac{E}{4} = 2 \quad m$
    - At R : $E_k = \frac{1}{2}m2^2 = 2 \quad m = E_p \quad lost$
- Therefore, when the mass is at R, it has half the final speed and so has gained a kinetic energy of $\frac{E}{4}$, this is when it has lost a potential energy of $\frac{E}{4}$

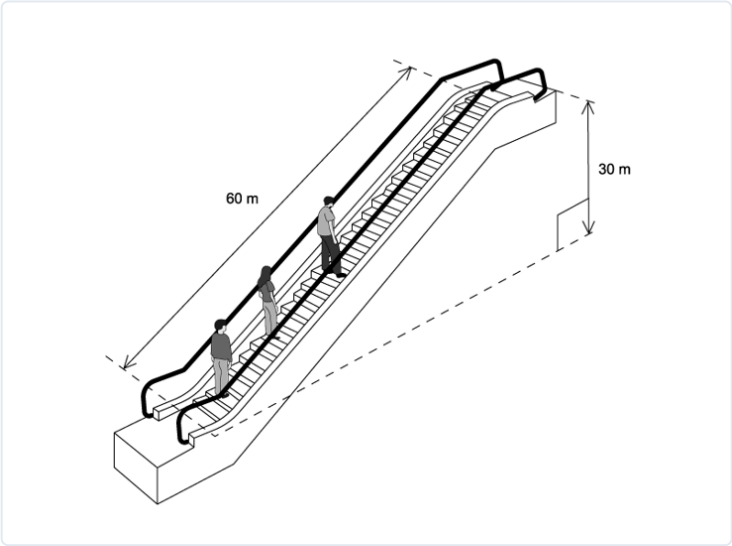
A is incorrect because At Q : $E_k = \frac{1}{2}m1^2 = 0.5m = E_p \quad lost$, this is a loss of $\frac{E}{16}$

C is incorrect because At S :$E_k = \frac{1}{2}m3^2 = \frac{9}{2} = E_p \quad lost$, this is a loss of $\frac{16E}{9}$

D is incorrect because At R : $KE = \frac{1}{2}m2^2 = 2m = PE \quad lost$, this is a loss of $\frac{E}{4}$

Question 26 Hard [1 mark]

An escalator is 60 m long and lifts passengers through a vertical height of 30 m, as shown.



To drive the escalator against the forces of friction when there are no passengers requires a power of 2.0 kW.

The escalator is used by passengers of average mass 60 kg and the power to overcome friction remains constant.

How much power is required to drive the escalator when it is carrying 20 passengers and is travelling at 0.75 m s^{-1}?

A 4.4 kW

B 6.4 kW

C 8.8 kW

D 10.8 kW

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Total weight of passengers = W = 20 mg
- Each passenger is moved a distance of 60 m as the escalator goes up
- The weight of the passengers acts vertically downwards
- There will be a component of weight acting down the slope of the escalator
- Component of weight acting parallel to the slope of the escalator
    - Where θ is the angle the escalator makes with the horizontal

- $\sin\ \theta = \frac{O}{H} = \frac{30}{60} = 0.5$
- The component of the weight force acting parallel to the slope is mg sinθ or 0.5mg

- Work needs to be done against forces. Additionally, work should be done against the component of weight acting down the slope
- $P = \frac{E}{t} = \frac{W}{t} = \frac{Fd}{t} = F\frac{d}{t} = Fv$
- $P = 2000 + Fv$
- $P = 2000 + 20\ mgv\ \sin\ \theta$
- $P = 2000 + 20 \times 60 \times 9.81 \times 0.50.75$

  $= 6414.5\ \text{W} = 6.4\ \text{kW}$

Therefore, the power required by the escalator is 6.4 kW

Exam tip: many calculation questions you will come across have multiple ways in which you can solve them. Another way you could solve this question would be to consider the gain in potential energy of the 20 passengers, try it out! Which method do you find easier ?

Question 27 Hard [1 mark]

A wind turbine has blades that sweep an area of 2000 m^2. It converts the power available in the wind to electrical power with an efficiency of 50 %.

What is the electrical power generated if the wind speed is 10 m s^{-1}? (The density of air is 1.3 kg m^{-3})

A 130 kW

B 650 kW

C 1300 kW

D 2600 kW

Mark Scheme & Explanation:

*The correct answer is **B** because:*

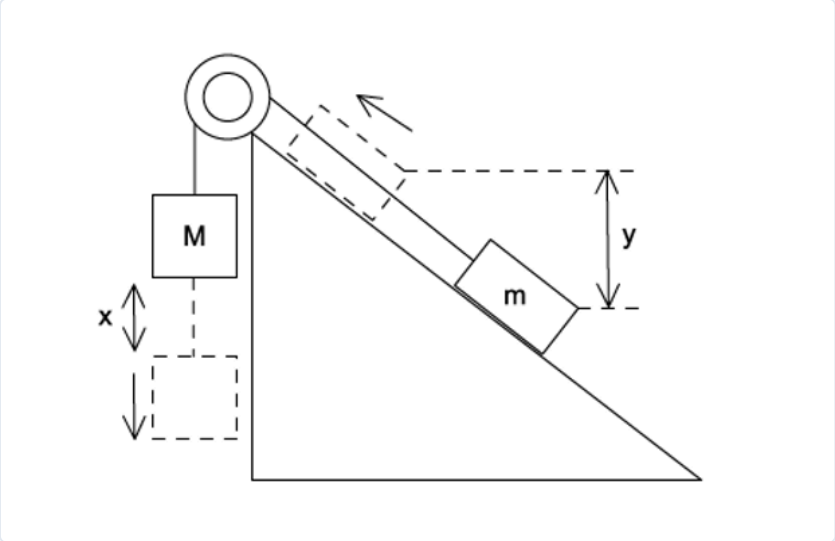
- The kinetic energy of the air causes the wind turbine to rotate which is then converted to electrical power
  - Kinetic energy of air = $\frac{1}{2}mv^2$
- The speed is known so we need to find the mass using:
  - Density = $\frac{mass}{volume}$, so, mass = density × volume
  - Density of air = 1.3 kg m^{-3}
- The wind speed is 10 m s^{-1} and this wind blows on an area of 2000 m^2
- This means in one second, a column of air of 10 m in length and with an area of 2000 m^2 moves past the blades of the wind turbine
  - This is equivalent to a volume of 10 × 2000 = 20 000 m^3 s^{-1}
- The mass of air passing the blades per second = density × volume of air per second
- Mass of air passing the blades per second : $\frac{m}{t} = 1.3 \times 20\ \ 000 = 26\ \ 000\ \ kg\ \ s^{-1}$
  - Power from wind = $\frac{energy}{wind} = \frac{mv^2}{2t} = \frac{1}{2}\frac{m}{t}v^2$
- Efficiency = 50%, so, only 50% of the power of the wind is converted to electrical power
  - Electrical Power = $0.50 \times 0.5 \times 26\ \ 000 \times 10^2 = 650\ \ 000 = 650\ \ kW$

Therefore, the electrical power output of the turbine is 650 kW

Question 28 Hard [1 mark]

The figure shows a mass m on an inclined plane connected by an inextensible string over a smooth pulley to a larger mass *M*. When the mass *M* falls through a distance *x*, the mass *m* moves up a vertical distance *y* at a constant speed.

The friction between *m* and the inclined plane is *F*.



What amount of energy is converted to heat?

A $F \quad x - y$

B Fy

C $Mgx - mgy$

D $M - m \quad gx$

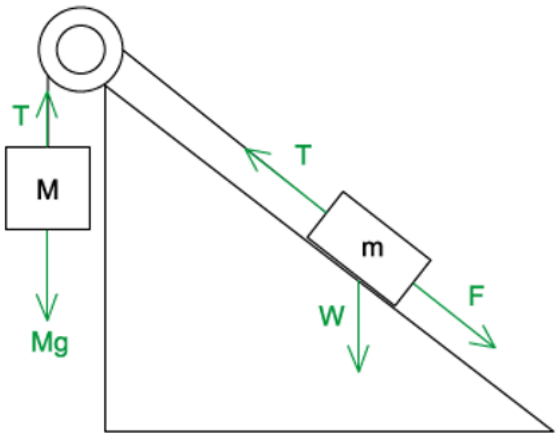
Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Work done = Force applied × displacement in the direction of force
- The question asks how much energy is converted to heat energy; this energy comes from the friction experienced by mass *m* moving up the rough plane
- In this case the forces acting on the mass are its weight W, the frictional force F and the tension T in the string holding the mass which is equal to the weight of mass *M*, so we can write
    - Vertical force on mass *M* : $T = Mg$

- Weight of mass m : $W = mg$
- Vertical forces on mass m : $F + W = T$
- Therefore, the frictional force is : $F = T$ - $W = Mg$ - mg

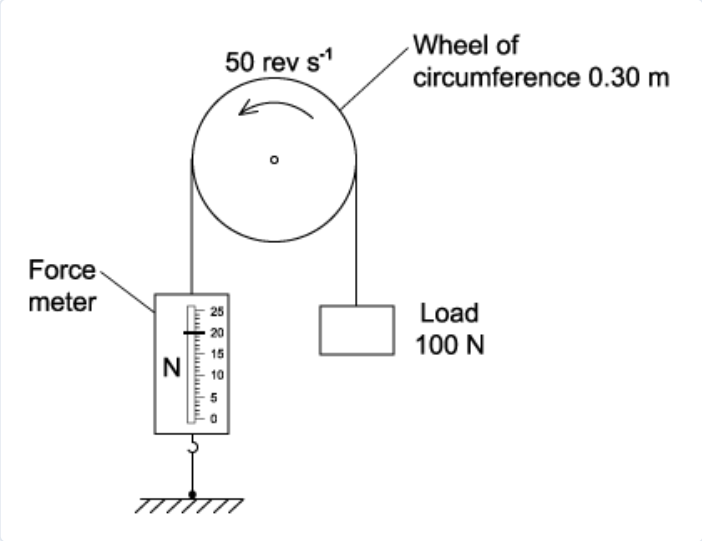
- Vertical displacement by mass M = x
- Work done by mass M = Mgx
- Vertical displacement by mass $m = y$
- Work done by mass m $m = mgy$
- Amount of energy converted to heat : Mgx - mgy



Question 29 Hard [1 mark]

The diagram shows a wheel of circumference 0.30 m. A rope is fastened at one end to a force meter. The rope passes over the wheel and supports a freely hanging load of 100 N. The wheel is driven by an electric motor at a constant rate of 50 revolutions per second.

When the wheel is turning at this rate, the force meter reads 20 N.



What is the output power of the motor?

A 0.3 kW

B 1.2 kW

C 1.8 kW

D 3.8 kW

Mark Scheme & Explanation:

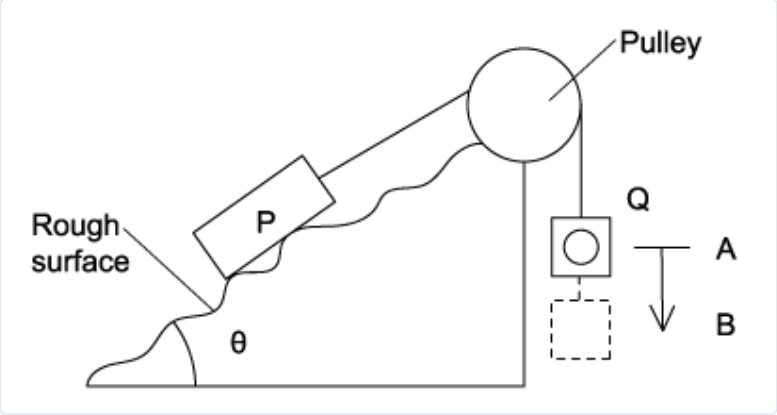
*The correct answer is **B** because:*

- The force due to the 100 N load causes the right end of the rope to accelerate downwards and simultaneously causes the left end to move upwards
- The force due to the 20 N on the force meter causes the left end to accelerate downwards and simultaneously causes the right end to move upwards
- Even if both the 100 N and the 20 N act downwards, they have opposite effects on the rope and the direction of turning of the wheel

- Since the 100 N load is greater than the 20 N force from the force meter, there is a resultant force acting downwards on the load. This force would cause the wheel to turn in a clockwise direction if not balanced by the motor
- To keep the load from moving, the wheel should also exert a force equal in magnitude and in opposite direction to the rope
    - From Newton's 3rd law since the resultant force of the rope is 100 – 20 = 80 N downwards
- So, the force exerted by the wheel on the rope is 80 N upwards
- The wheel turns in an anticlockwise direction at a rate of 50 revolutions per second
- This means the speed of the motor is 50 revolutions per second, and one revolution is equal to the circumference of the wheel, which is 0.3 m
- So, the distance moved against the force in one second, or its speed is $= 50 \times 0.30 = 15 \quad \text{m} \quad \text{s}^{-1}$
    - Power provided by motor $= Fv = 80 \times 15 = 1200 \quad \text{W} = 1.2 \quad \text{kW}$
- Therefore, the power provided by the motor is 1.2 kW

Question 30 Hard [1 mark]

The figure below shows two blocks P and Q, initially stationary and connected to each other by a taut string passing over a smooth pulley. The system is released from rest and block Q moves vertically downwards from position A to B.



Which of the following statements is correct?

A Potential energy of the block Q at A = Kinetic energy of block Q at B

B Loss in potential energy of the whole system = Gain in kinetic energy of the whole system + work done against friction due to block P moving against a rough surface

C Loss in potential energy of the whole system = Gain in kinetic energy of the system

D Loss in potential energy of block Q = Gain in potential energy of block P

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- From the principle of conservation of energy:
  - The sum of energies of blocks P and Q when Q is at position B should be equal to the initial amount of gravitational potential energy (GPE) the blocks possess when Q is at position A
- When the two blocks are at rest, they are above the ground so they both possess GPE
- After Q is released, it begins to move down and transfers GPE to kinetic energy (KE) as it increases in speed, this pulls block P up the slope with the same speed and hence the same KE
- We are told the surface of the slope is rough so block P does some work against friction
- Overall, the GPE at the beginning transfers to the KE of the blocks minus the heat loss due to friction on the slope, this can be written:

- Loss in potential energy of the whole system = Gain in kinetic energy of the whole system + work done against friction due to block P moving against a rough surface

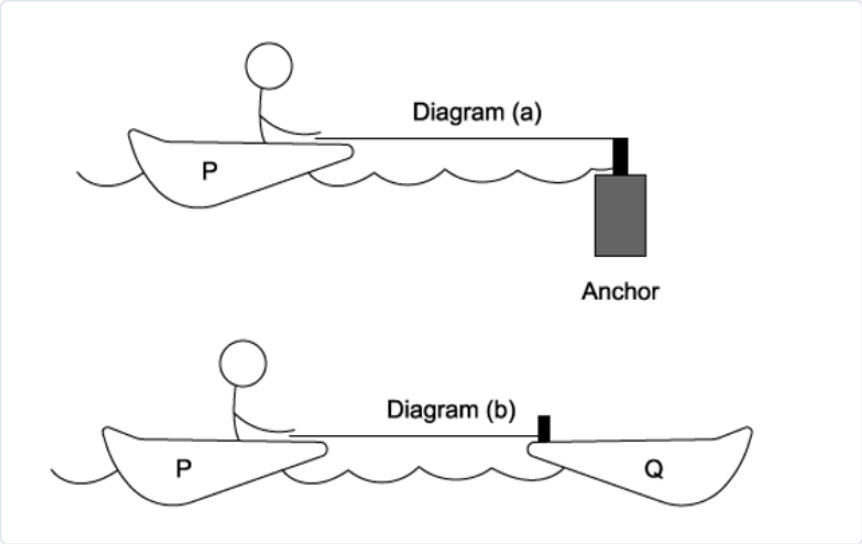
A is incorrect because this statement is true for block Q if it was not attached to block Q, but since it is you cannot ignore the energy transfers of block P

C is incorrect because this statement ignores the work done against friction as the slope is rough

D is incorrect because the loss in potential energy of the whole system is equal to (the loss in potential energy of block Q) – (gain in potential energy of block P) but this statement does not take into account the KE gained or the work done due to friction so it is incorrect

Question 31 Hard [1 mark]

A boy on a boat P pulls on a rope with a constant force *F* over a duration of time *t*. The other end of the rope is either tied to an anchor on the pier (Diagram a) or a freely floating boat Q (Diagram b) of equal mass as shown in the figures shown.



W_a and W_b represent the total work done by the boy during the time *t* and P_a and P_b represent the average power output of the boy in diagrams (a) and (b) respectively.

Which of the following statements is correct?

A $W_a > W_b \quad and \quad P_a > P_b$

B $W_a = W_b \quad and \quad P_a = P_b$

C $W_a < W_b \quad and \quad P_a < P_b$

D $W_a < W_b \quad and \quad P_a = P_b$

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Power $= \frac{work\ \ done}{time\ \ taken} = \frac{force \times displacement}{time\ \ taken}$
- Since the anchor is fixed to the pier, the boy will not be able to move it, regardless of how much force he applies, he can only move the boat towards the pier
- Hence, the work done and power will be close to zero in Diagram (a)
- For the freely floating boat Q, there is a higher chance that the boy will be able to move himself and boat Q for a distance

- A higher force will be required by the boy to travel the same distance as in Diagram (a)
- Hence, more work is done and a power output substantially larger than zero will be required in Diagram (b)
- Therefore, both work done W_b and power output P_b will be higher than work done W_a and power output P_a
    - $W_a < W_b \quad and \quad P_a < P_b$

A is incorrect because this is statement is telling us that it requires less energy and power output to pull the boat towards the pier than boat Q

B & D are incorrect because since the mass of the pier and boat Q vary greatly, the work done and power outputs to pull boat P towards them cannot be the same

Question 32 Hard [1 mark]

What is the power required to give a body of mass *m* a forward acceleration *a* when it is moving with velocity *v* up a frictionless track inclined at an angle θ to the horizontal?

A $mavg \sin \theta$

B $mav \sin \theta + mgv$

C $mav + mgv \sin \theta$

D $mav + mgv \sin \theta$

Mark Scheme & Explanation:

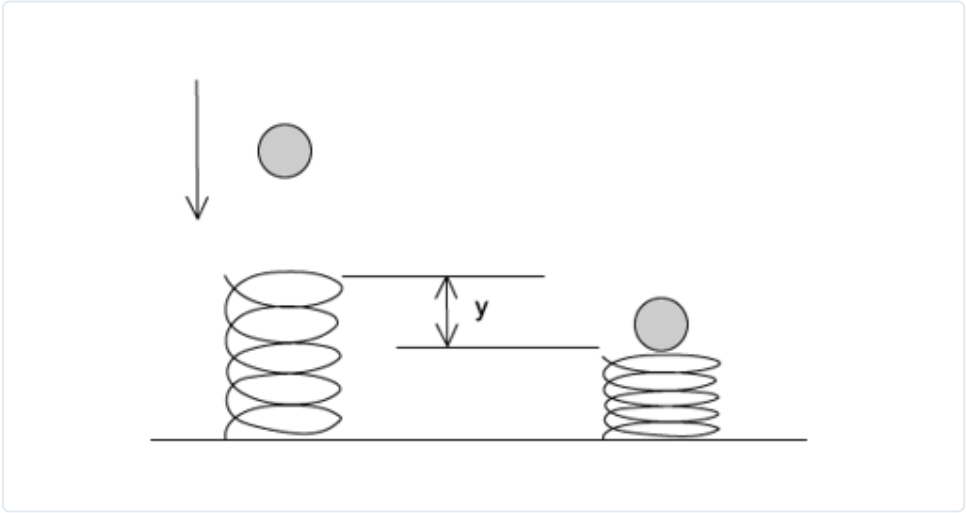
*The correct answer is **C** because:*

- Power $\frac{work\ done}{time\ taken} = \frac{force \times distance}{time\ taken} = force \times velocity$
- The track is frictionless, so, no frictional force acts on the mass
- If the track was horizontal the power required to give the mass *m* a forward acceleration *a* when it is moving with velocity *v* is given by
  - Power $P = Fv = ma \times v = mav$ **(1)**
- But in this case, the track is inclined at an angle θ to the horizontal
- The component of the weight of the mass (against motion) along the track is equal to $= mg \sin \theta$
- Therefore, a force is required to keep the mass stationary and prevent it from moving down along the inclined track
- This force should be equal in magnitude to the component of the weight so that the resultant acceleration (when considering only the component of the weight and this force) is zero
- This force would result in a power equal to
  - Power $P = mg \sin \theta \times v = mgv \sin \theta$ **(2)**
- Total power required is given by adding Equation **(1)** and Equation **(2)**:
  - Total power = $mav + mgv \sin \theta$

Question 33 Hard [1 mark]

A ball of mass *m* falls freely from rest. When it has reached a speed *v*, it strikes a vertical spring.

The spring is compressed by a distance *y* before the ball moves upwards again.



Assume that all energy the ball loses becomes elastic potential energy in the spring.

What is the average force exerted by the spring during its compression?

A $\frac{mv^2}{2y}$

B $\frac{m}{2y}v^2 - 2gy$

C $\frac{mv^2}{y}$

D $\frac{m}{2y}v^2 + 2gy$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Let's consider the motion of the ball from the point of impact with the spring to the maximum compression of the spring
- From the conservation of energy :
- Loss in kinetic energy (KE) + gravitational potential energy (GPE) = work done to compress ball = gain in elastic potential energy in spring
- Loss in ball's KE $= \frac{1}{2}mv^2$
- Gain in GPE = mgy

- Work done by ball = $Average \quad force \times displacement = F_{av}y$
  - $\frac{1}{2}mv^2$ - $0 + mgy = F_{av}y$
  - $F_{av} = \frac{\frac{1}{2}mv^2 + mgy}{y}$
  - $F_{av} = \frac{m}{2y}v^2 + 2gy$

Question 34 Hard [1 mark]

A hydroelectric power station has a turbine with blades that sweep through water flowing at a rate of 15 m^3 s^{-1}. The water has a density of 1000 kg m^{-3} and flows at a speed of 8.0 m s^{-1}. The turbine converts the kinetic energy of the flowing water to electrical energy with an efficiency of 40%.

What is the electrical power output of the turbine?

A 96 kW

B 190 kW

C 380 kW

D 480 kW

Mark Scheme & Explanation:

The correct answer is B

- The kinetic energy of the flowing water causes the turbine to rotate, which is then converted to electrical power
- Kinetic energy is given by

$$E_k = \frac{1}{2}mv^2$$

- The speed is known, so to find the mass flow rate:

Mass flow rate = density × volume flow rate

- Where
  - density of water = 1000 kg m^{-3}
  - volume flow rate = 15 m^3 s^{-1}

Mass flow rate = 1000 × 15 = 15 000 kg s^{-1}

- Power from flowing water is

$$P = \frac{\text{energy}}{\text{time}} = \frac{mv^2}{2t} = \frac{1}{2}\frac{m}{t}v^2$$

- Efficiency = 40%, so only 40% of the power of the flowing water is converted to electrical power
- Electrical power is

$$P = 0.40 \times 0.5 \times 15000 \times 8.0^2 = 192\ \ 000\ \text{W} = 192\ \text{kW}$$

$$P \approx 190\,\text{kW} \quad 2 \text{ s.f.}$$